**MIDTERM EXAMINATION**

**SEMESTER FALL 2004**

**MTH401- Differential Equations**

**Total Marks: 50**

**Duration: 60min**

## Instructions

**1. Attempt all questions.**
**2. The Time allowed for this paper is 60 minutes.**
**3. This examination is closed book, closed notes, closed neighbors; any one found cheating will get zero grades in the course MTH401 Differential Equations.**
**4. You are not allowed to use any type of Table for Formulae of Differentiation and integration during your exam.**
**5. Each objective type question carries 2 marks and each Descriptive question carries 10 marks. So write every step of the solution of descriptive question to get maximum marks.**
**6. Do not ask any questions about the contents of this examination from anyone. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.**



The differential equation $\frac{dy}{dx} - y \quad e^{x} y^{2}$ is not Bernoulli equation.

T

F

$f(x, y) \quad \frac{x}{x^{2} \quad y^{2}}$ Is homogeneous.

T

F

Population dynamics are not practical application of the first order differential equations.

T

F

A set

$y_1, y_2, \ldots, y_n$

Of $n$ linearly dependent solutions, on interval $I$, of the homogeneous linear $n$th-order differential equation

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = 0$$

Is said to be a fundamental set of solutions on the interval $I$.

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**F**

The differential operator that annihilates $10x^3 - 2x$ is $D^4$.

**T**

F

**(a)** Define separable form. Just separate the variable of the given differential equation.

$$(3r\theta - 3\theta + r - 1)\,dr - (2r\theta - 4\theta + r - 2)\,d\theta = 0$$

The differential equation of the form d y / d x = f (x, y) is called **separable** if it can be written in the form d y / d x = h(x)g(y)

$$(3r\theta - 3\theta + r - 1)\,dr - (2r\theta - 4\theta + r - 2)\,d\theta = 0$$

$$(3r\theta - 3\theta + r - 1)\,dr = (2r\theta - 4\theta + r - 2)\,d\theta$$

$$[3\theta(r-1) + 1(r-1)]\,dr = [2\theta(r-2) + 1(r-2)]\,d\theta$$

$$(r-1)(3\theta+1)\,dr = (r-2)(2\theta+1)\,d\theta$$

$$\frac{r-1}{r-2}\,dr = \frac{2\theta+1}{3\theta+1}\,d\theta$$

No: 6

**(b)** Check whether the given differential equation is exact or not if not then make it exact also show that it is exact **(Just make the equation exact do not solve it further)**.

$$\left(\frac{3y^2 - x^2}{y^5}\right)\frac{dy}{dx} + \frac{x}{2y^5} = 0$$

It can also be written as

$$xdx \quad 23 \quad y^2 - x^2 \quad dy \quad 0$$

$$M \quad xy \qquad x, N\, xy \quad , \qquad 6y^2 - 2x^2$$

$$M_y \quad 0, \quad N_x \quad -4x$$

$$M_y \neq N_x$$

Thus it is not exact now we apply techniques to make it exact

$$xdx \quad 6y^2 - 2x^2 \quad dy \quad 0$$

$$\frac{N_x - M_y}{M} \quad \frac{-4x}{x} \quad -4 \quad g(y)$$

$$IF. \quad e^{\int -4dy} \quad e^{-4y}$$

$$xe^{-4y}dx \quad e^{-4y} \quad 3y^2 - 2x^2 \quad dy \quad 0$$

$$M \quad xy \qquad xe^{-4y}, \quad N\, xy, \qquad e^{-4y} \quad 3y^2 - 2x^2$$

$$M_y - \quad 4xe^{-4y}, N_x - \quad 4xe^{-4y}$$

$$M_y \quad N_x$$

Which shows that equation is exact

**(a)** Solve the Bernoulli equation $x^3 \frac{dy}{dx} \quad 2xy \quad y^5$ ( **Just make the given equation linear in v, do not integrate** )

$$x^3 \frac{dy}{dx} \quad 2xy \quad y^5$$

$$\frac{dy}{dx} y^{-5} \quad \frac{2}{x^2} y^{-4} \quad \frac{1}{x^3}$$

$$put\, y^{-4} \quad v$$

$$-4 \frac{dy}{dx} y^{-5} \quad \frac{dv}{dx}$$

$$\frac{dy}{dx} y^{-5} - \quad \frac{1}{4} \frac{dv}{dx}$$

*Then*

$$-\frac{1}{4} \frac{dv}{dx} \quad \frac{2v}{x^2} \quad \frac{1}{x^3}$$

$$\frac{dv}{dx} - \frac{8v}{x^2} \quad \frac{-4}{x^3}$$

Thus it is linear in “v”.

**(b)**

Initially there were 200 milligrams of a radioactive substance present. After 8 hours the mass increased by 4%. If the rate of decay is proportional to the amount of the substance present at any time, determine half-life of the radioactive substance? **(Just make the model of the radioactive decay as well as describe the given conditions do not solve further)**

Suppose that $A_0$ is the initial amount, as given $A_0$ is 200 and A(t) be the amount present at time t then its governed by the differential equation

$$\frac{dA}{dt} \alpha A$$

$$\frac{dA}{dt} kA$$

$$\frac{dA}{A} kdt$$

Integrate both sides

$$\ln Aktc$$

$$Ae^{kt c}$$

$$Aee^{kt c}$$

$$A \quad A_0e^{kt} \qquad say \; A_0 \quad e^c$$

Where $A_0$ is the initial amount

$A_0$=200 = A(0)

$$A(8)=200+\frac{4}{100}\;200$$

A (8)=208

A (T) =100

**(a)**

Find a second solution of following differential equations where the first solution is given **(also write the formulae).**

$$12 \quad x \quad y''2'4x0y,- \qquad y_1 \quad e^{-2x}$$

$$12 \quad x \quad y''2'4x0y,- \qquad y_1 \quad e^{-2x}$$

differential equation can be written as

$$y'' \frac{2x}{12 \quad x} y' - \frac{4}{12 \quad x} \quad 0$$

the 2nd solution is given by

$$y_2 \quad y_1 \int \frac{e^{-\int p dx}}{y_1^{2}} dx$$

$$y_2 \quad e^{-2x} \int \frac{e^{-\int \frac{2x}{12\ \ x} dx}}{e^{-2x\ \ 2}} dx$$

$$y_2 \quad e^{-2x} \int \frac{e^{-\int \frac{12\ 1x-}{12\ \ x} dx}}{e^{-4x}} dx$$

$$y_2 \quad e^{-2x} \int \frac{e^{\int (\frac{1}{12\ \ x} -1) dx}}{e^{-4x}} dx$$

$$y_2 \quad e^{-2x} \int \frac{e e^{-x} \ \ln(1\ 2)x}{e^{-4x}} dx$$

$$y_2 \quad e^{-2x} \int \frac{e^{2\ln(1\ 2)x}}{e^{-3x}} dx$$

$$y_2 \quad e^{-2x} \int \frac{e^{\ln(1\ 2)x}}{e^{-3x}} dx$$

$$y_2 \quad e^{-2x} \int (1 \quad 2x) e^{3x} dx$$

$$y_2 \quad e^{-2x} \quad (1 \quad 2x) \frac{e^{3x}}{3} - \int (1 \quad 2x)' \frac{e^{3x}}{3} dx$$

$$y_2 \quad e^{-2x} \left[ (1 \quad 2x) \frac{e^{3x}}{3} - \int \frac{2e^{3x}}{3} dx \right]$$

$$y_2 \quad \left[ (1 \quad 2x) \frac{e^{x}}{3} - \frac{2e^{x}}{9} \right]$$

**(b)**

Solve the differential equation by the undetermined coefficient **(superposition approach)**

$y'' - 2y' - 3y \quad e^{4x}$

If complimentary solution is given below

$y_c \quad c_1 e^{3x} \quad c_2 e^{-x}$

Then just find particular solution.

We find a particular solution of non-homogeneous differential equation.

Suppose input function

$y_p \quad Ae^{4x}$

Then

$y'_p \quad 4Ae^{4x}$

$y''_p \quad 16Ae^{4x}$

Substituting in the given differential equation

$16Ae^{4x} - 2\,4\;Ae^{4x} \quad -3Ae^{4x} \quad e^{4x}$

$e^{4x}\;16A - 8A - 3A \qquad e^{4x}$

From the resulting equations

$5A \quad 1$

$A \quad \frac{1}{5}$

$y_p \quad \frac{1}{5}e^{4x}$

**(a)** Solve differential equation by the undetermined coefficient **(annihilator operator).**

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} \quad x\cos x$

If complimentary solution is given below

$y_c \quad c_1 \quad c_2e^{4x}$

Then just find general solution.

In this case of input function is

g(x)= $x\cos x$

further

$(D^2 \quad 1)^2 g(x) \quad (D^2 \quad 1)^2 (x\cos x) \quad 0$

Therefore the differential operator $(D^2 \quad 1)^2$ annihilates the function g. operating on both sides

$(D^2 \quad 1)^2 (D^2 - 4D)\,y \quad (D^2 \quad 1)^2 (x\cos x)$

$(D^2 \quad 1)^2 (D^2 - 4D)\,y \quad 0$

This is the homogeneous equation of order 6. Next we solve this higher order equation.

Thus auxiliary equation is

$(m^2 \quad 1)^2(m^2-4m)\,y \quad 0$

$m(m-4)(m^2 \quad 1)^2 \quad 0$

$m \quad 0,4,i,i\;-i,-i$

Thus its general solution of the differential equation must be

$y \quad c_1 \quad c_2e^{4x} \quad (c_3 \quad c_4x)\cos x \quad (c_5 \quad c_6x)\sin x$

**(b)**

Solve the differential equation by the variation of parameters

$y''-y \quad x^2$

Complimentary solution is given below If

$y_c \quad c_1e^x \quad c_2e^{-x}$

Then just find particular solution **(do not integrate).**

$y''-y \quad x^2$

This equation is already in standard form

y′′+ P(x)y′ +Q(x)y = f(x)

Therefore, we identify the function f(x) as

f(x) = $x^2$

Since $y_1 \quad e^x,\; y_2 \quad e^{-x}$ so

We construct the determinants

$$W(y_1, y_2) \quad \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} \quad -e^{x-x}-e^{x-x} \quad -2$$

$$W_1 \quad \begin{vmatrix} 0 & e^{-x} \\ x^2 & -e^{-x} \end{vmatrix} \quad -x^2e^{-x}$$

$$W_2 \quad \begin{vmatrix} e^x & 0 \\ e^x & x^2 \end{vmatrix} \quad x^2e^x$$

We determine the derivatives of the function $u_1$ and $u_2$

$$u'_1 \quad \frac{W_1}{W} \quad \frac{-x^2e^{-x}}{2} \rightarrow u_1 \quad \int \frac{x^2}{2}e^{-x}dx$$

$$u'_2 \quad \frac{W_2}{W} \quad \frac{x^2e^{-x}}{2} \rightarrow u_2 \quad -\int \frac{x^2}{2}e^{-x}dx$$

$y_p \quad u_1e^x \quad u_2e^{-x}$

is required particular solution